FROM PIXELS TO PERCEPTION

What if AI Could See Your UI?

White on white.
Ghost button.
Fully invisible.

All tests passed.

But the user couldn’t find the save button.

Our tools were looking at code.

Nobody was looking at the screen.

Suriya Prakash

Hello!

Principal Software Engineer
User Experience Engineering

Red Hat, India

The bug wasn't in the logic.

It was in the perception.

TODAY'S JOURNEY

01 What is Visual Testing

→

02 Why It Matters

→

03 When It Goes Wrong

→

04 The AI Solution

Let's go.

WHAT IS VISUAL TESTING?

Does what users see match what we intended?

WHY IT MATTERS

50ms

That’s how long users take to form an opinion.

Before they read a word. Before they click anything. Just from looking.

You don’t just have a bug.

You have a trust problem.

WHEN VISUAL DESIGN GOES WRONG

APPLE MAPS

Mapgate.

- Warped 3D buildings, wrong landmarks, missing roads
- Visual testing would have caught it before release.

SNAPCHAT

3M+
users lost

£700M
market cap wiped

- Stories feed restructured overnight.
- A layout diff test would have flagged this before it shipped.

THE PROBLEM WITH HOW WE TEST TODAY

Challenges of Traditional Visual Testing

01 Manual process

Time-consuming. Error-prone.

02 Subjective Results

What looks broken to one tester looks fine to another.

03 Scalability Issues

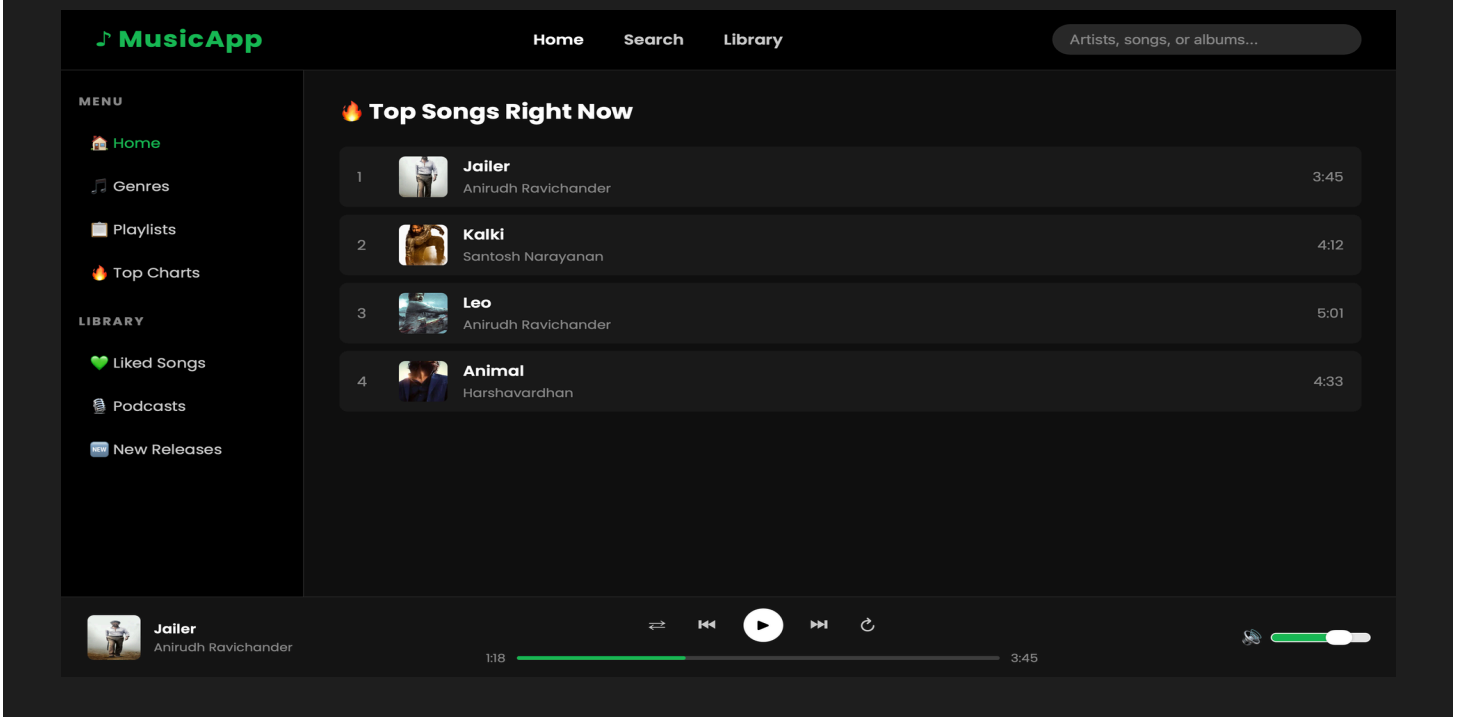
Multiplies testing effort exponentially.

04 Subtle Differences

1px shifts. Faded colours. The eye misses what a diff tool catches.

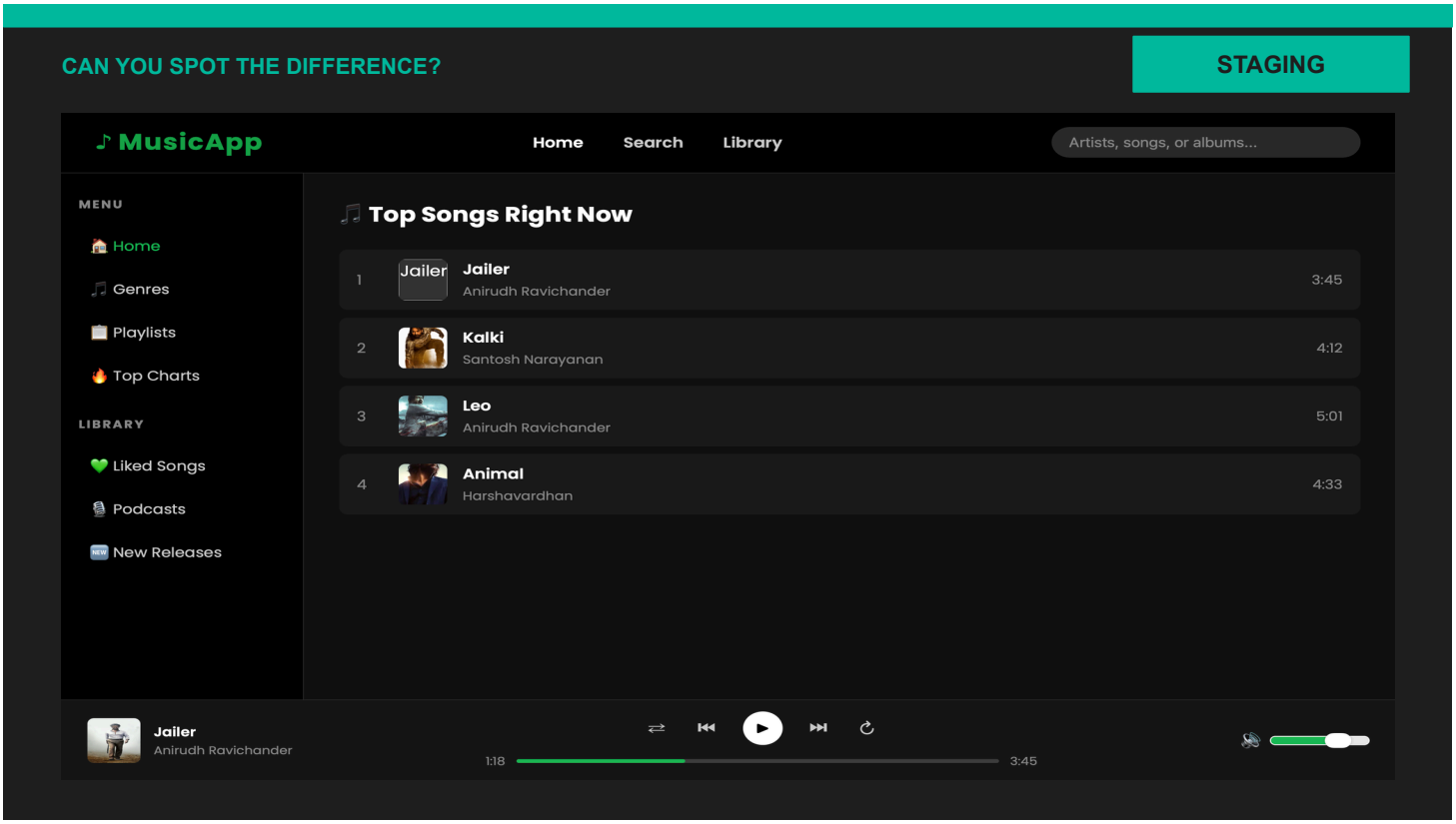
CAN YOU SPOT THE DIFFERENCE?

PRODUCTION

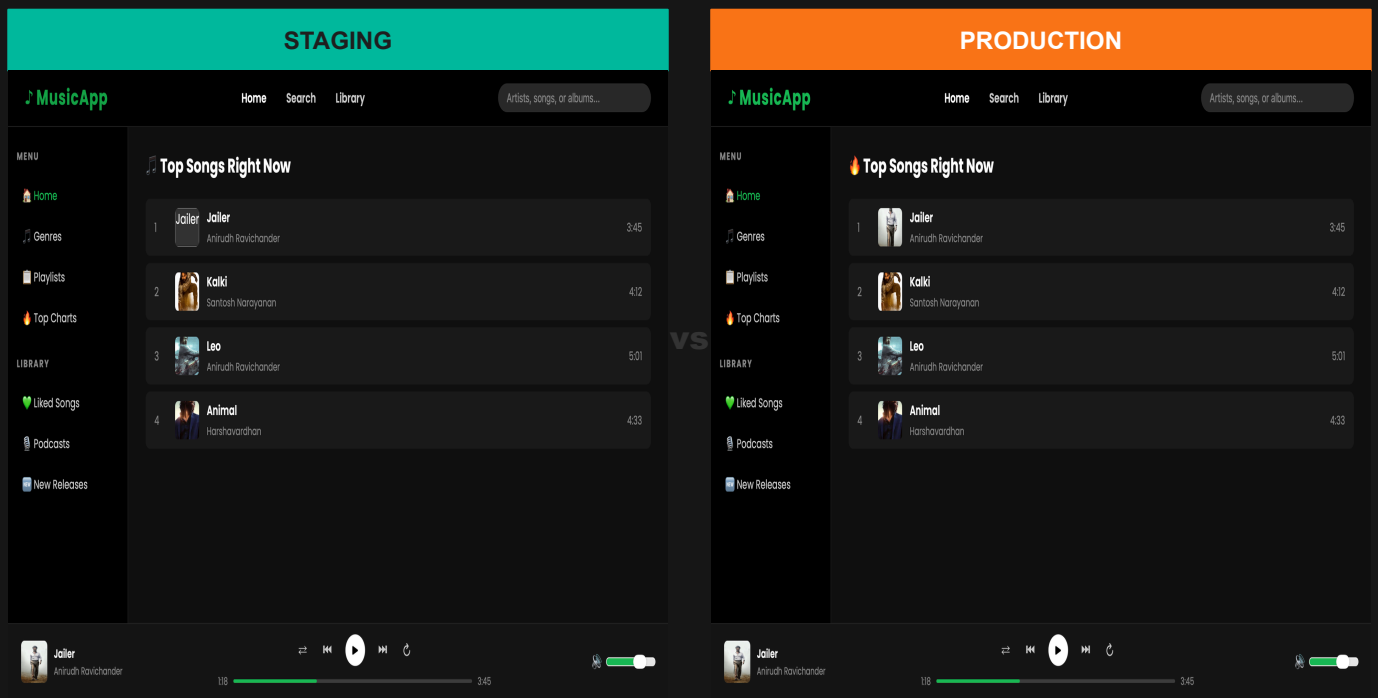


CAN YOU SPOT THE DIFFERENCE?

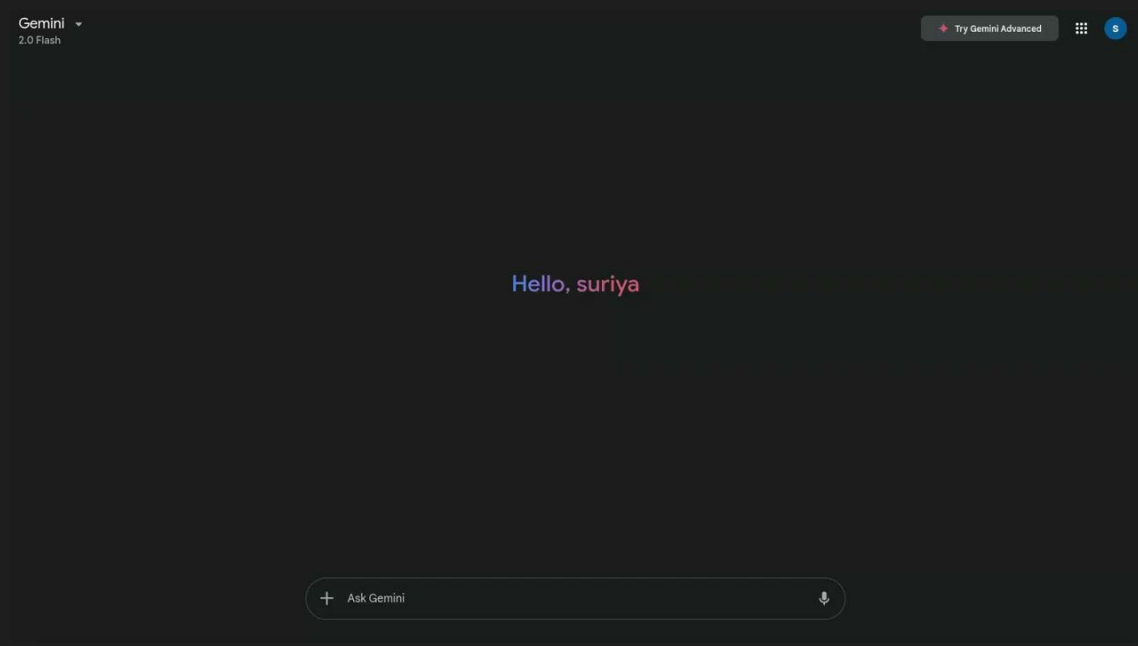
STAGING



SPOT THE DIFFERENCE

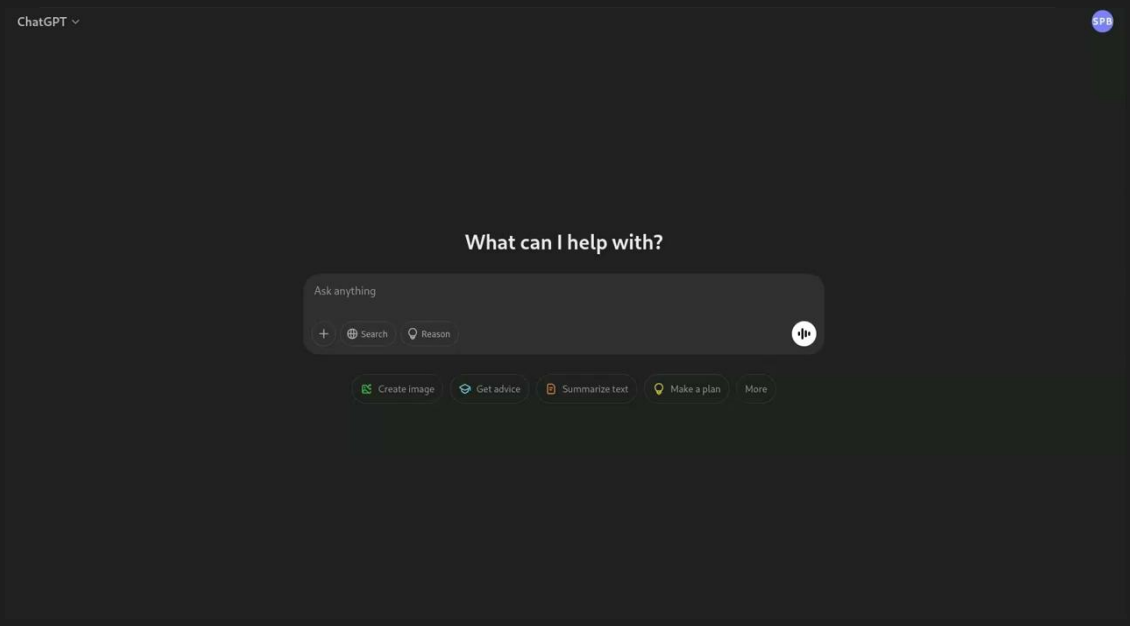


HUMAN EYE VS AI — ROUND 1



Can it spot what changed?

HUMAN EYE VS AI — ROUND 2



Same question. Different model. What changes?

What changes when AI sees your UI

BEFORE		AFTER
✖ Manual screenshot reviews	VS	✔ AI catches every pixel diff
✖ Different results each tester		✔ Same baseline every run
✖ 1px shifts go unnoticed		✔ Cross-browser in one run
✖ Bugs shipped on Friday		✔ CI/CD blocks the bad deploy
✖ Expensive tools just to see a diff		✔ Free to run forever

PROOF OF CONCEPT

Redefining Visual Testing with AI

FRAMEWORK	LANGUAGE	AI MODEL
Playwright	**Python**	**ResNet50**
Test Automation	Core scripting language	Deep learning model

ALGORITHM	LIBRARY	LIBRARY
SSIM	**OpenCV**	**NumPy**
Structural Similarity Index	Computer vision library	Numerical computation

Examples of Popular Vision Models

Microsoft **ResNet-50** Microsoft Research	Google **EfficientNet** Google	VGG **VGG-16** Oxford / VGGNet
Meta **DINOv2** Meta AI	Google **MobileNetV3** Google	Microsoft **Swin Transformer** Microsoft Research
Meta **ConvNeXt** Meta AI	**CLIP** OpenAI	Microsoft **EfficientViT** MIT / Microsoft

Why ResNet-50 + SSIM?

Decision rationale

ResNet-50

- Lightweight & Fast
- Pretrained Weights Available
- Proven Feature Extractor
- Production-Ready & Stable

SSIM (Structural Similarity)

- Human-Aligned Metric
- Complements ResNet Features
- No Training Required
- Interpretable Score

HOW IT WORKS

The Visual Testing Pipeline

1 Cypress | 2 OpenCV | 3 NumPy | 4 SSIM | 5 ResNet50

SSIM Detects the Difference

Pixel level change detected.

DEFECT FLAGGED

ResNet50 Detects the Difference

Feature level change detected.

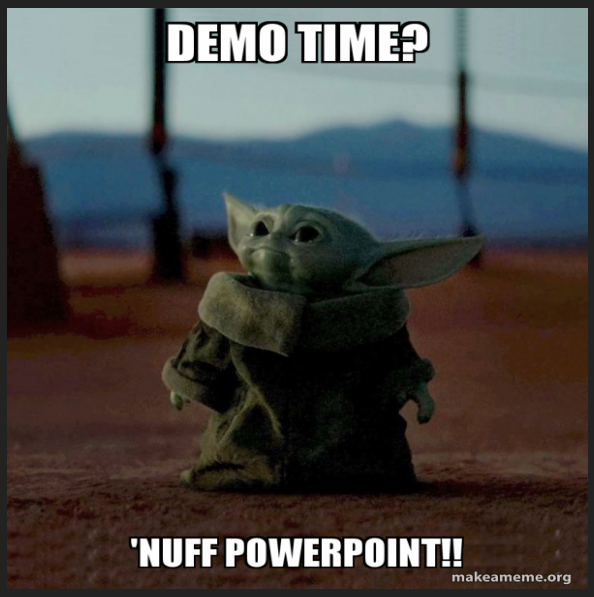
DEFECT FLAGGED

PATH C

Both Pass

SSIM ≥ 0.99 and ResNet feature diff = 0. No visual regression detected.

PASS



Test the pixels.

Trust the perception.

THANK YOU

in suriyaprakash19

gh suriya-93

[] suriya_prakash:matrix.org